# Lunar solar-TES sCO2 power system: summary

Run: 2026-11-06 14:31
Demand 100 kWe | TES 800 -> 450 K | compressor inlet 320-380 K at 8.0 MPa | P_high 25 MPa

| | Simple | Recuperated | |
|---|---|---|---|
| Cycle efficiency at full charge | 15.9 | 33.4 | % |
| TES mass | 3,380 | 2,144 | t |
| TES volume | 2,254 | 1,430 | m3 |
| Collector (mirror) area | 1,927 | 1,059 | m2 |
| Plant radiator panel area | 1,128 | 1,284 | m2 |
| Plant radiator mass | 15.8 | 18.0 | t |
| PHX size UA | 71.9 | 66.0 | kW/K |
| TES lowest in operation | 548 | 577 | K |
| Parasitic share of gross output | 15.7 | 10.4 | % |
| Demand served (energy) | 100.0 | 100.0 | % |
| Light on receiver -> electricity | 11.2 | 20.5 | % |

Status lines from the report:
Status (simple): ALL OK
Status (recuperated): ALL OK

Contents: summary, run log, full report, figures, references.

## Run log (messages from Steps 2-14)

```
Inputs OK: cycles ['simple', 'recuperated'], TES 800 -> 450 K, TIT max 790 K, PR 3.12, 100 kWe + parasitic (3 kWe fixed, 0
        kWe day), compressor inlet 320-380 K at 8.0 MPa, day 354 h / night 354 h
Building simple-cycle map (25 x 91 points)...
simple cycle: turbine 24.75 -> 8.08 MPa, eta at TIT 790 K / T1 320 K = 15.94 %, usable TES down to 475 K
Building recuperated-cycle map (25 x 91 points)...
recuperated cycle: turbine 24.50 -> 8.16 MPa, eta at TIT 790 K / T1 320 K = 33.42 %, usable TES down to 480 K
     simple: PHX UA   71.9 kW/K (design: TES 475 K, 23.15 kg/s, CO2 in 413 K, effectiveness 0.854)
recuperated: PHX UA   66.0 kW/K (design: TES 480 K, 20.46 kg/s, CO2 in 412 K, effectiveness 0.866)
     simple: usable TES down to 476 K with the PHX (cycle alone: 475 K)
recuperated: usable TES down to 480 K with the PHX (cycle alone: 480 K)
     simple: radiator installed   1128 m2 of panel, 2-sided (15.8 t at 14 kg/m2)
recuperated: radiator installed   1284 m2 of panel, 2-sided (18.0 t at 14 kg/m2)
100 kWe, PR = 3.12, regolith = highlands

               ---- Simple ----          -- Recuperated --
 T_TES    eta     Q_in      TIT      eta     Q_in      TIT
    K      %       kW        K        %       kW        K
   800   15.9      680      800     33.9      320      800
   775   15.9      681      775     32.7      332      775
   750   15.9      683      750     31.3      346      750
   725   15.8      685      725     29.9      363      725
   700   15.7      689      700     28.4      382      700
   675   15.6      695      675     26.7      406      675
   650   15.4      703      650     24.9      436      650
   625   15.2      714      625     22.9      473      625
   600   14.8      731      600     20.7      523      600
   575   14.3      756      575     18.3      593      575
   550   13.6      796      550     15.6      697      550
   525   12.5      864      525     12.5      871      525
   500   10.8     1006      500      9.9     1090      500
   475               cannot run                cannot run
   450               cannot run                cannot run
Usable TES window (simple): 800 K down to 476 K
Usable TES window (recuperated): 800 K down to 480 K
Receiver: alpha 0.95, emissivity 0.85, concentration 500x, T_rec = T_TES + 50 K | optics 0.839 x cosine (0.866 at noon, 0.707
         at the horizon)
 sun elev   TES 450 K   TES 650 K   TES 800 K
      90°       94.4%       92.7%       89.9%
      45°       94.4%       92.5%       89.6%
      30°       94.3%       92.4%       89.4%
      15°       94.3%       92.3%       89.1%
      10°       94.3%       92.3%       89.0%
       5°       94.3%       92.2%       88.9%
Radiator sink: night 3 K, noon 3 K (shielded)
     simple: TES  3380.4 t ( 2254 m3) | collector  1927 m2 | curtailed  0.0% | survives True | repeats to 0.000 K
recuperated: TES  2144.3 t ( 1430 m3) | collector  1059 m2 | curtailed  0.0% | survives True | repeats to 0.000 K
     simple: TES balance -3.38e-10 kWh, cycle balance 3.41e-13 kW
recuperated: TES balance 4.55e-12 kWh, cycle balance 3.41e-13 kW
Step 13: 10 figures prepared for the PDF
Step 14: 19 references and 11 assumptions listed for the PDF
```

# Report (Step 12)

```
================================================================================
  LUNAR POWER SYSTEM REPORT
  Solar tower -> regolith TES -> sCO2 Brayton cycle: simple vs recuperated
================================================================================

[1] INPUTS (shared)
--------------------------------------------------------------------------------
  Lunar day / night                                  354 / 354 h
  Time step                                           60 min
  Electrical demand / generator eff.             100.0 / 0.95 kWe
  Parasitic: controls / heliostat drives    3.0 kWe / 0.5 W/m2
  Charging loop: dp / dT / circulator eff.  2 % / 150 K / 0.70
  TES temperature range                          800 -> 450 K
  Regolith type / density                  highlands / 1500 kg/m3
  Regolith cp at T_max / T_min                   1114 / 936 J/kg-K
  TES mass margin                                   1.200
  TES insulation / ground temperature          1.00 m / 250 K
  Compressor inlet: lowest / highest              320 / 380 K
  Compressor inlet / outlet pressure          8.00 / 25.00 MPa
  Pressure drops: PHX / recup / cooler         1.0 / 1.0 / 1.0 %
  Compressor / turbine efficiency                 0.85 / 0.90
  Recuperator effectiveness                          0.95
  PHX: approach at design / design TES      10 K / 475 , 480 K
  Solar: G / optics (rho x dust x intercept)   1361 W/m2 / 0.839
  Heliostat cosine factor: noon / horizon        0.866 / 0.707
  Receiver: absorptance / emissivity              0.95 / 0.85
  Receiver: concentration / T above TES           500x / 50 K
  Collector mode / latitude                 heliostat / 0 deg
  Habitat: crew / volume / wall area       4 / 419 m3 / 336 m2
  Habitat: indoor T / MLI emittance              295 K / 0.020
  Habitat: share of 100 kWe used inside                1.00
  Habitat radiator / heat-pump hot side           283 / 360 K
  Radiator: emissivity / area margin              0.85 / 1.00
  Radiator view to surface / shield                0.50 / yes
  Surface temperature night / noon                  100 / 392 K

[2] CYCLE STATES AT DESIGN POINT (TES fully charged, compressor inlet 320 K)
--------------------------------------------------------------------------------
  Simple cycle
    State                         T (K)   P (MPa)
    1  Compressor inlet           320.0      8.00
    2  Compressor outlet          413.1     25.00
    3  Turbine inlet              790.0     24.75
    4  Turbine outlet             660.4      8.08
  Recuperated cycle
    State                         T (K)   P (MPa)
    1  Compressor inlet           320.0      8.00
    2  Compressor outlet          413.1     25.00
    3  Recuperator cold out       602.7     24.75
    4  Turbine inlet              790.0     24.50
    5  Turbine outlet             662.6      8.16
    6  Cooler inlet               425.1      8.08

[3] DESIGN-POINT PERFORMANCE
--------------------------------------------------------------------------------
                                                Simple       Recuperated
  Thermal efficiency                             15.94             33.42  %
  Carnot limit                                    59.5              59.5  %
  Turbine pressure ratio                          3.06              3.00
  Back-work ratio (W_c / W_t)                    0.417             0.423
  Electricity to habitat                         100.0             100.0  kWe
  Parasitic loads (night)                          3.0               3.0  kWe
  Gross generator output                         103.0             103.0  kWe
  Cycle shaft power (net)                        108.4             108.4  kW
  CO2 mass flow                                   1.32              1.35  kg/s
  Turbine power                                  184.0             186.2  kW
  Compressor power                                75.6              77.7  kW
  Recuperator duty                                 0.0             379.3  kW
  Heat from TES                                  680.0             320.0  kW
  Cooler heat rejection                          571.6             211.6  kW
  Cooler inlet temperature                       669.0             425.6  K

[4] SIZING
--------------------------------------------------------------------------------
                                                Simple       Recuperated
  Paper method m_TES (Eq. 21, 24)               2386.9            1123.4  t
  Night-only minimum m_TES                      2817.0            1786.9  t
  Design m_TES (with margin)                    3380.4            2144.3  t
  TES volume (Eq. 25)                             2254              1430  m3
  TES surface (cube)                              1031               761  m2
```

## Report (Step 12)  (continued)

```
  Collector area                                    1927              1059  m2
  Radiator area, largest night need                 1128              1284  m2
  Radiator panel area installed (fixed)             1128              1284  m2
    2-sided panels, mass at 14 kg/m2                15.8              18.0  t
  PHX size UA                                       71.9              66.0  kW/K
  Usable TES lower limit: cycle / with PHX   475 / 476         480 / 480  K

[5] OPERATION THROUGH THE LUNAR CYCLE
-------------------------------------------------------------------------------
                                                   Simple       Recuperated
  TES at sunrise                                    548.6             578.0  K
  TES lowest                                        547.8             577.0  K
    at (time after sunrise)                             1                 1  h
  TES at sunset                                     800.0             800.0  K
  Cycle efficiency, max (running)                    15.9              33.9  %
  Cycle efficiency, min (running)                    13.5              18.5  %
  CO2 mass flow, max                                 4.76              3.68  kg/s
  Turbine inlet, min-max                          548-800           577-800  K
  PHX gap (T_TES - TIT), min-max                  0.0-0.5           0.0-0.5  K
  Compressor inlet, night (min-max)               320-320           320-320  K
  Compressor inlet, day (min-max)                 320-320           320-320  K
  Hours compressor inlet above lowest                   0                 0  h
  Radiator use, max (running)                          43                27  %
  Hours at reduced power (radiator limit)               0                 0  h
    lowest power in those hours                         -                 -  kWe
  Hours cycle off                                       0                 0  h
  Demand served (energy)                            100.0             100.0  %

[6] ENERGY TOTALS PER LUNAR CYCLE (MWh)
-------------------------------------------------------------------------------
                                                   Simple       Recuperated
  Concentrated light on receiver                    629.87            346.13  MWh
  Solar heat to TES (net)                           578.79            318.14  MWh
  Solar curtailed (TES full)                          0.00              0.00  MWh
  Heat drawn by cycle                               576.50            316.38  MWh
    by day                                          326.77            175.85  MWh
    by night                                        249.74            140.53  MWh
  Gross electricity generated                        84.03             79.05  MWh
    parasitic loads                                  13.23              8.25  MWh
    electricity to habitat                           70.80             70.80  MWh
  Electricity NOT served                              0.00              0.00  MWh
  Waste heat                                        488.05            233.17  MWh
    to habitat                                        0.00              0.00  MWh
    to radiator                                     488.05            233.17  MWh
  TES heat loss (insulation)                          2.29              1.77  MWh
    as % of solar heat to TES                          0.4               0.6  %
    average / max                           3.2 / 5.3         2.5 / 3.9  kW
  Receiver loss (reflection + radiation)             51.08             27.99  MWh
  Receiver efficiency, lunar-day average             91.9              91.9  %
  Receiver efficiency, at noon                       92.2              92.3  %
  Daytime hours receiver closed (loss > input)                0                 0  h
    controls (always)                                2.12              2.12  MWh
    heliostat drives (day)                           0.34              0.19  MWh
    charging circulator (day)                       10.76              5.94  MWh
    peak parasitic load                              39.3              22.8  kWe
  Parasitic share of gross output                    15.7              10.4  %
  Light on receiver -> electricity                   11.24             20.45  %
  Curtailed fraction of solar heat                     0.0               0.0  %

[6b] HABITAT HEAT BALANCE (same for both cycles)
-------------------------------------------------------------------------------
  Wall heat loss, night (max)                          2.8 kW
  Wall heat gain, day (max)                            6.1 kW
  Heating needed from cycle, max                       0.0 kW
  Hours habitat needs heating                            0 h
  Cooling load, min / max                  97.8 / 106.8 kW
  Hours heat pump on                                     0 h
  Heat-pump power, max (inside 100 kWe)                0.0 kWe
  Habitat radiator panel area needed, max              173 m2
    mass at 14 kg/m2                                   2.4 t
                                                   Simple       Recuperated
  Heat sent to habitat from cycle                     0.00              0.00  MWh
  Habitat heating not met                             0.00              0.00  MWh
  Share of waste heat to habitat                       0.0               0.0  %

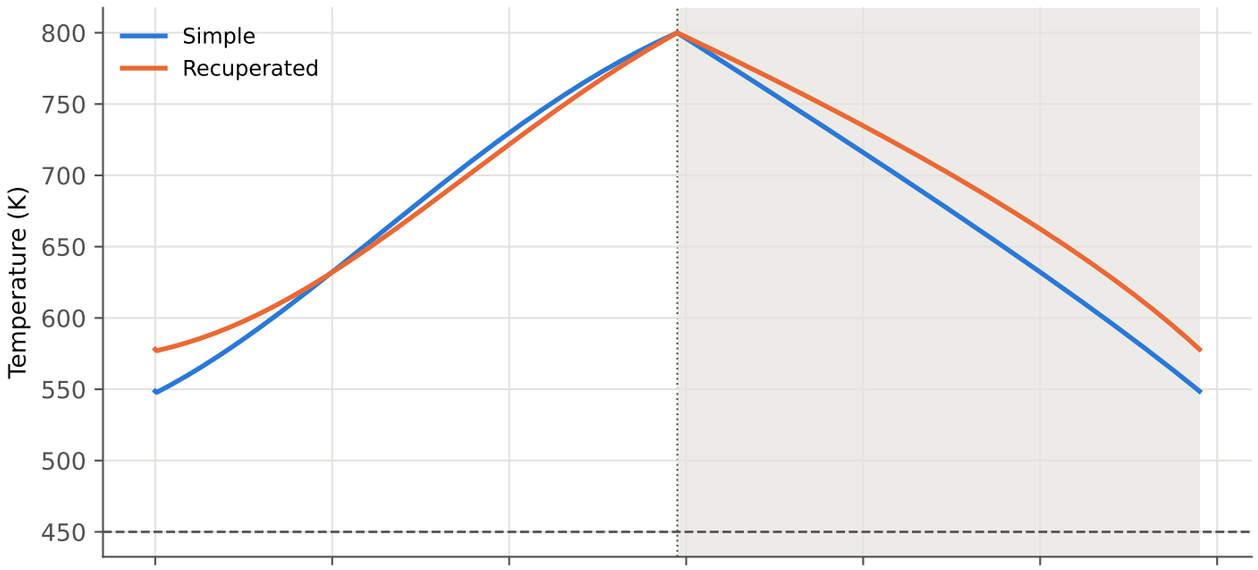
[7] CHECKS
-------------------------------------------------------------------------------
                                                   Simple       Recuperated
  Cycle energy balance residual                    3.4e-13           3.4e-13  kW
  TES energy balance residual                     -3.4e-10           4.5e-12  kWh
```

## Report (Step 12)  (continued)

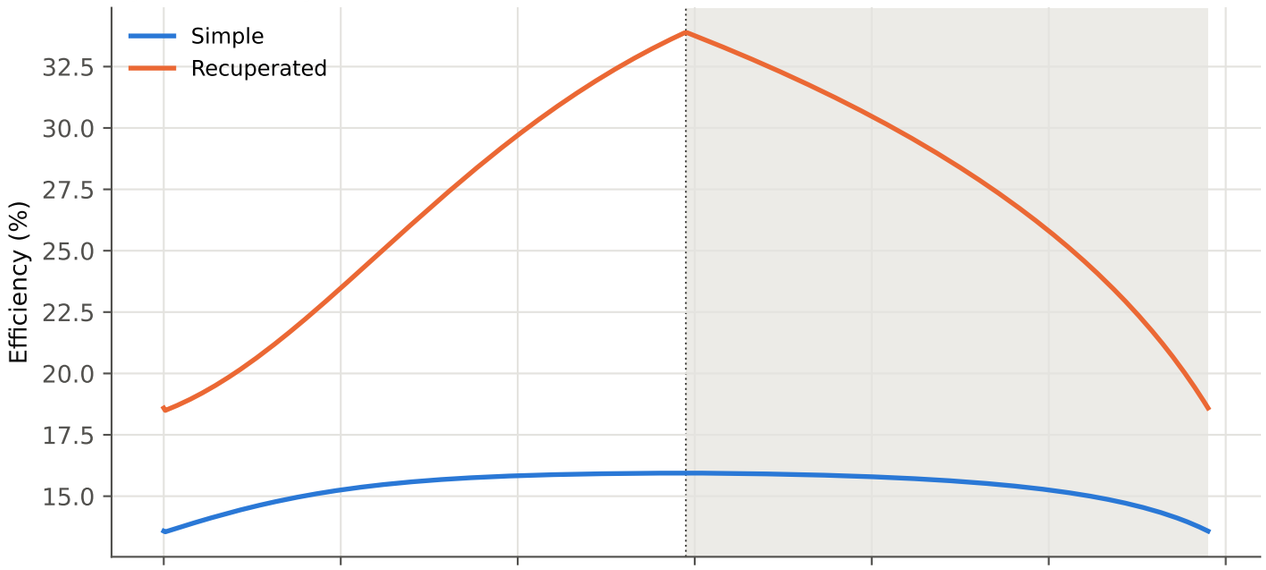
```
  Cycle repeats (T_TES change)                        -0.000            -0.000  K
  Status (simple): ALL OK
  Status (recuperated): ALL OK
================================================================================
```

Full lunar cycle, simple vs recuperated: Simple 3380 t TES / 1927 m2 collector, Recuperated 2144 t TES / 1059 m2 collector

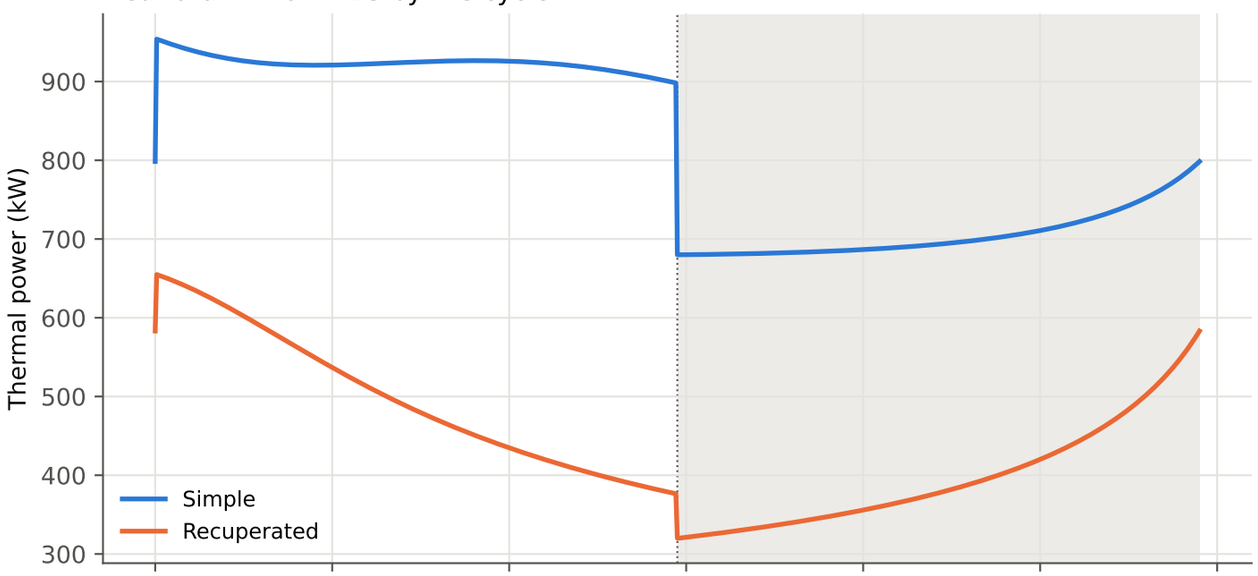
TES temperature



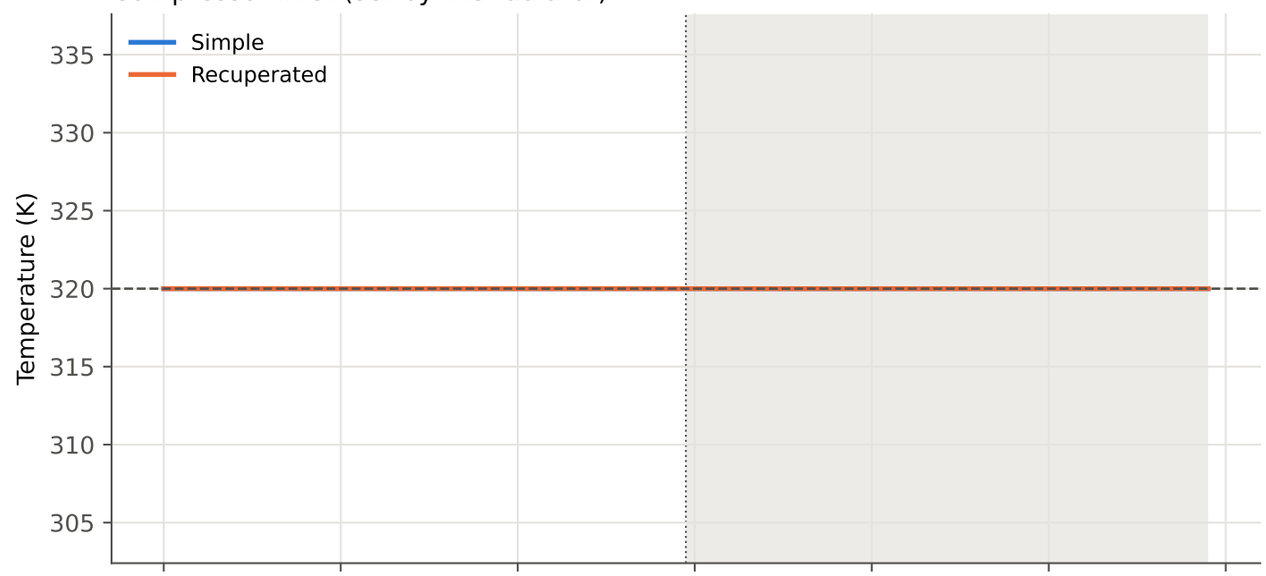

Cycle efficiency



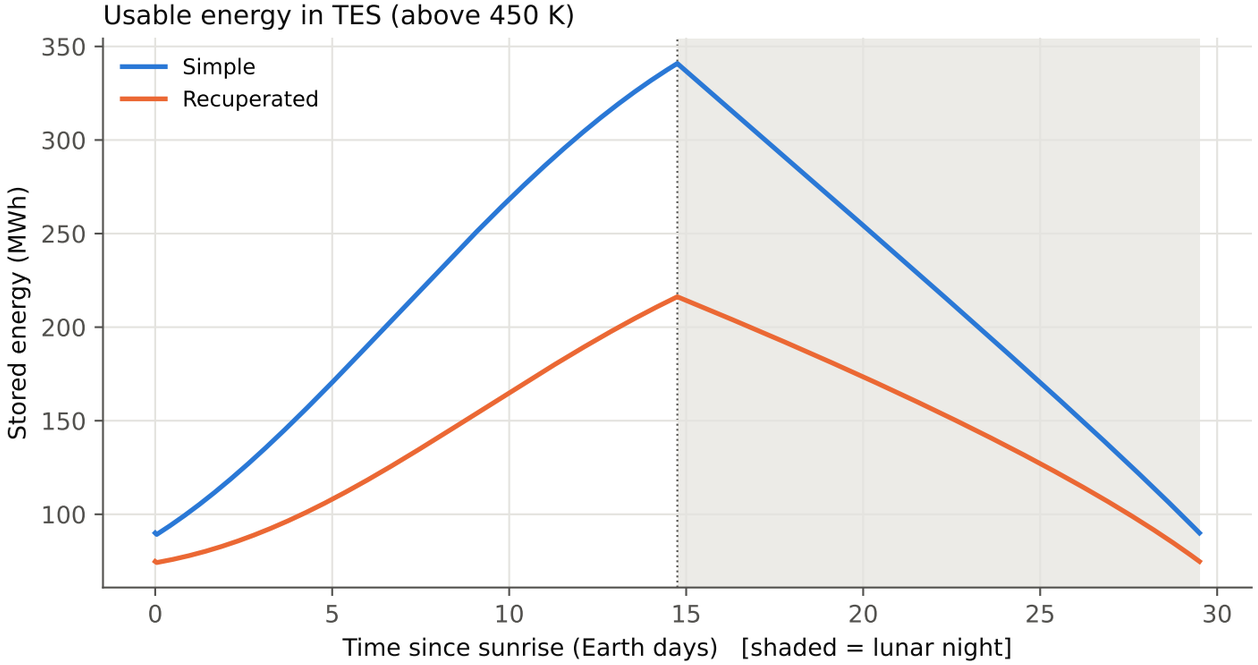

Heat drawn from TES by the cycle



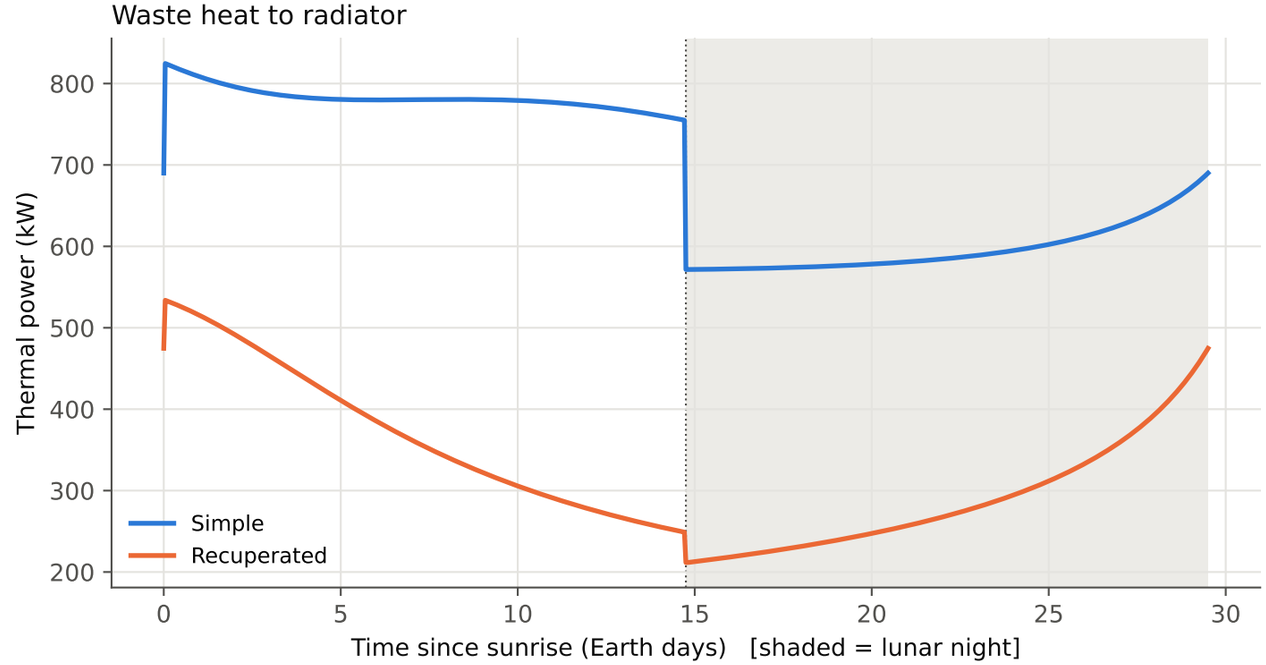

Compressor inlet (set by the radiator)

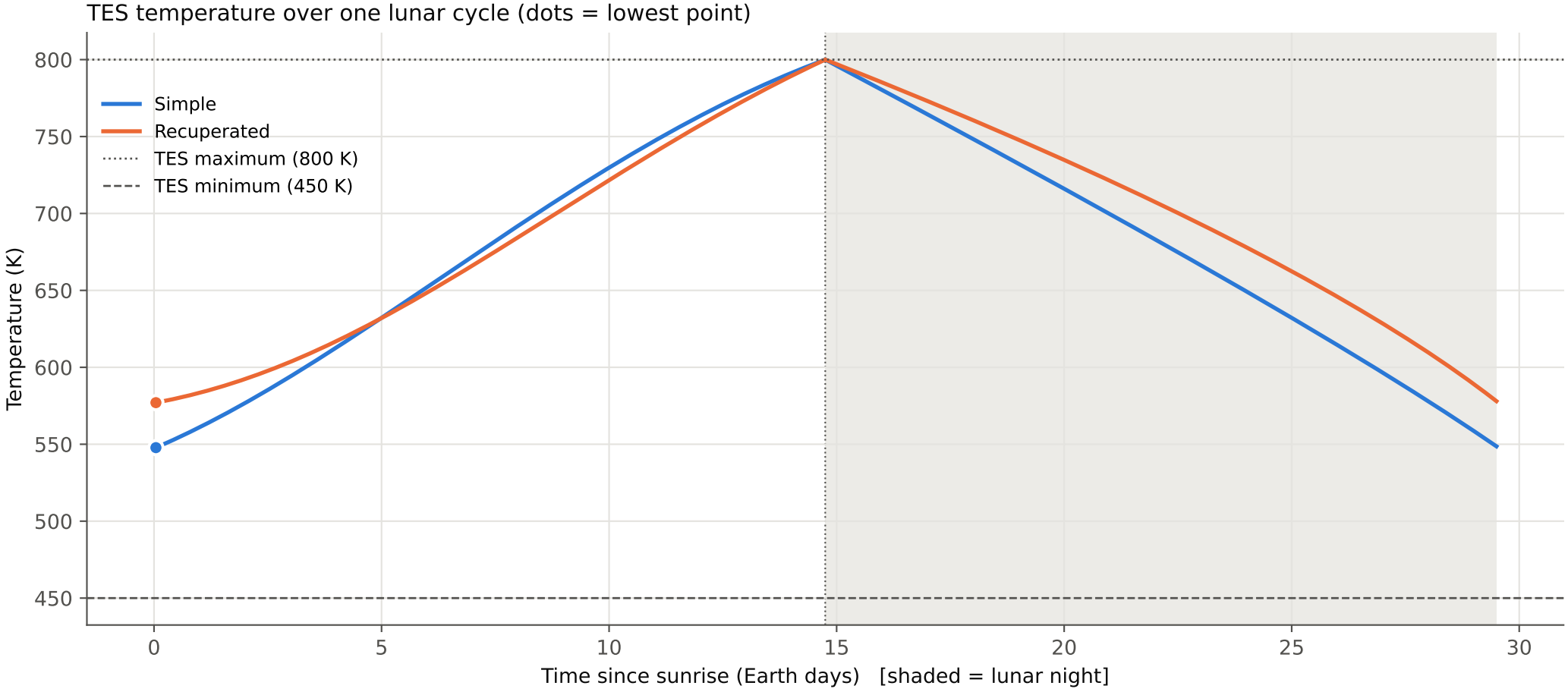



Usable energy in TES (above 450 K)

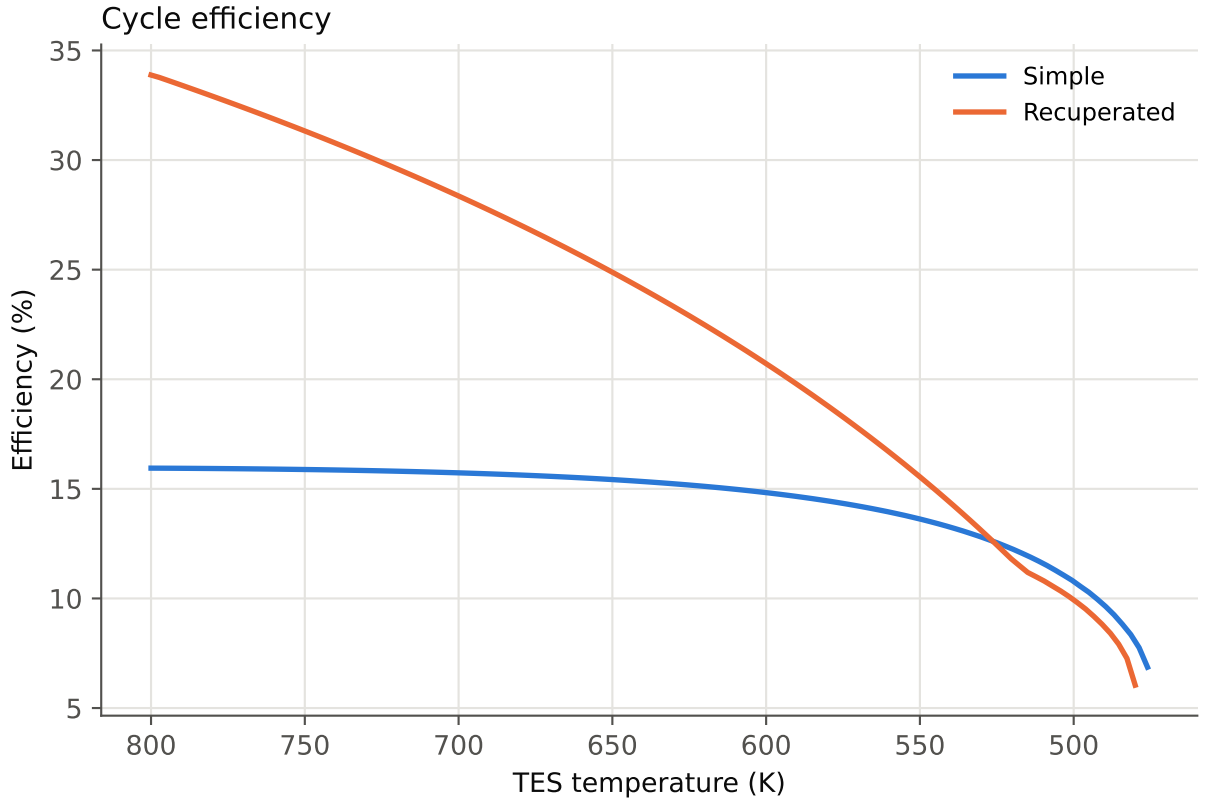


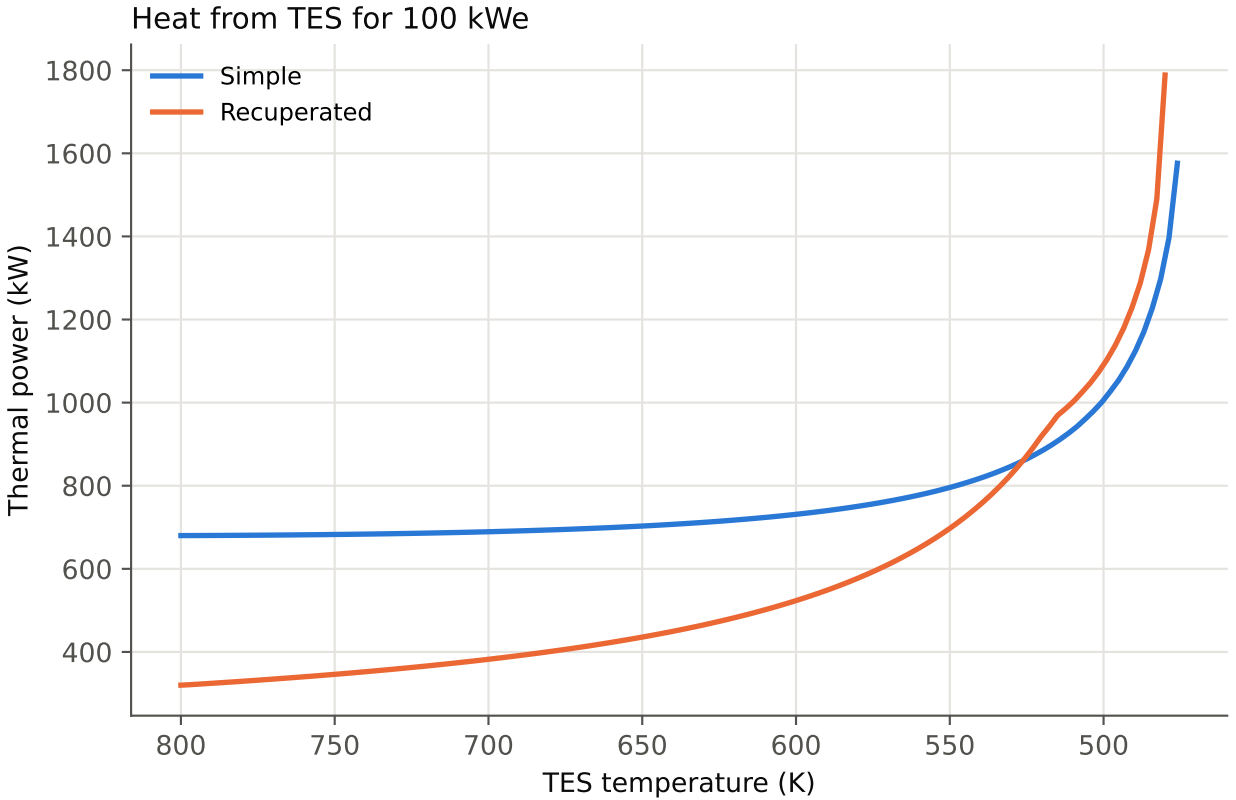

Waste heat to radiator

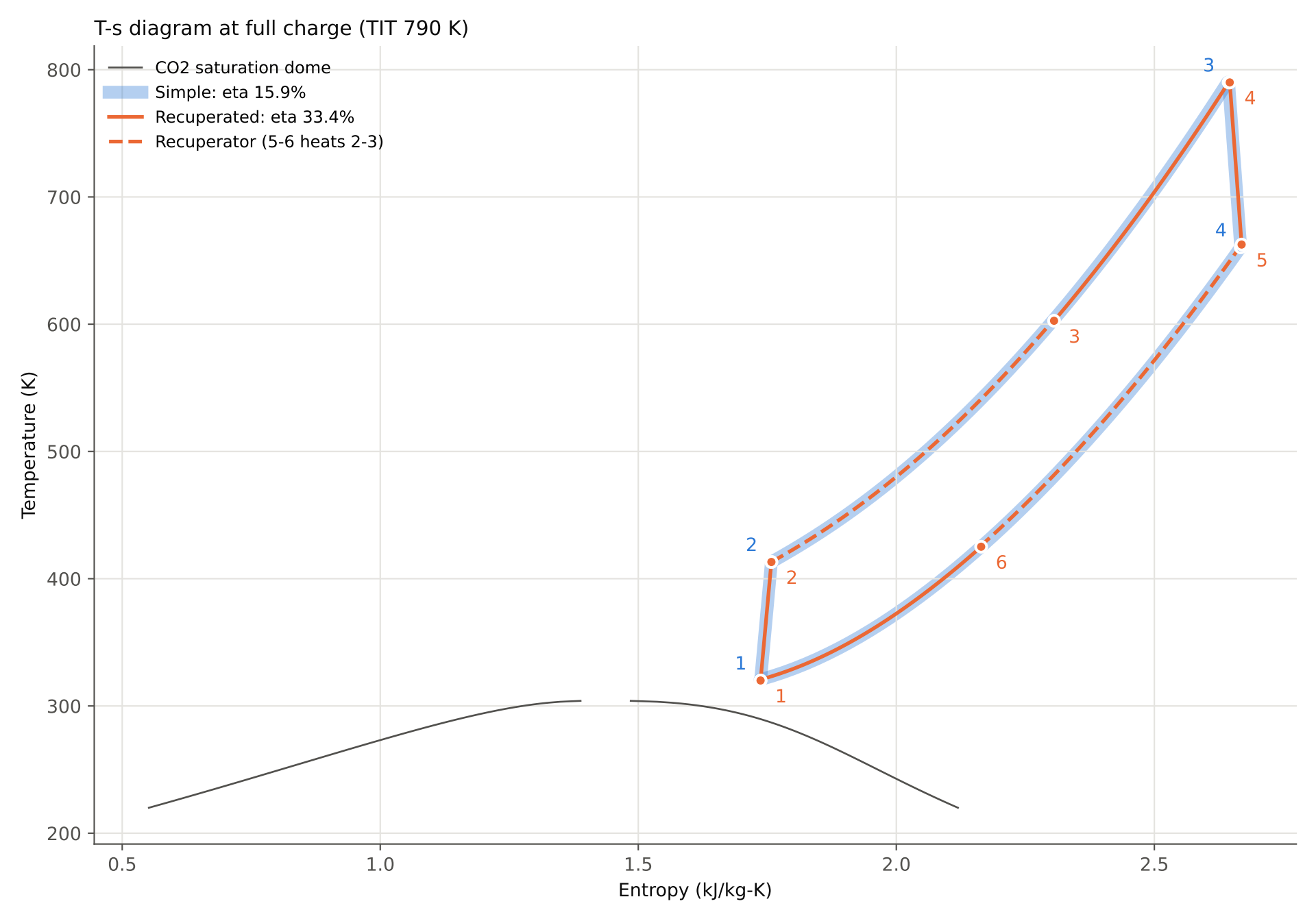


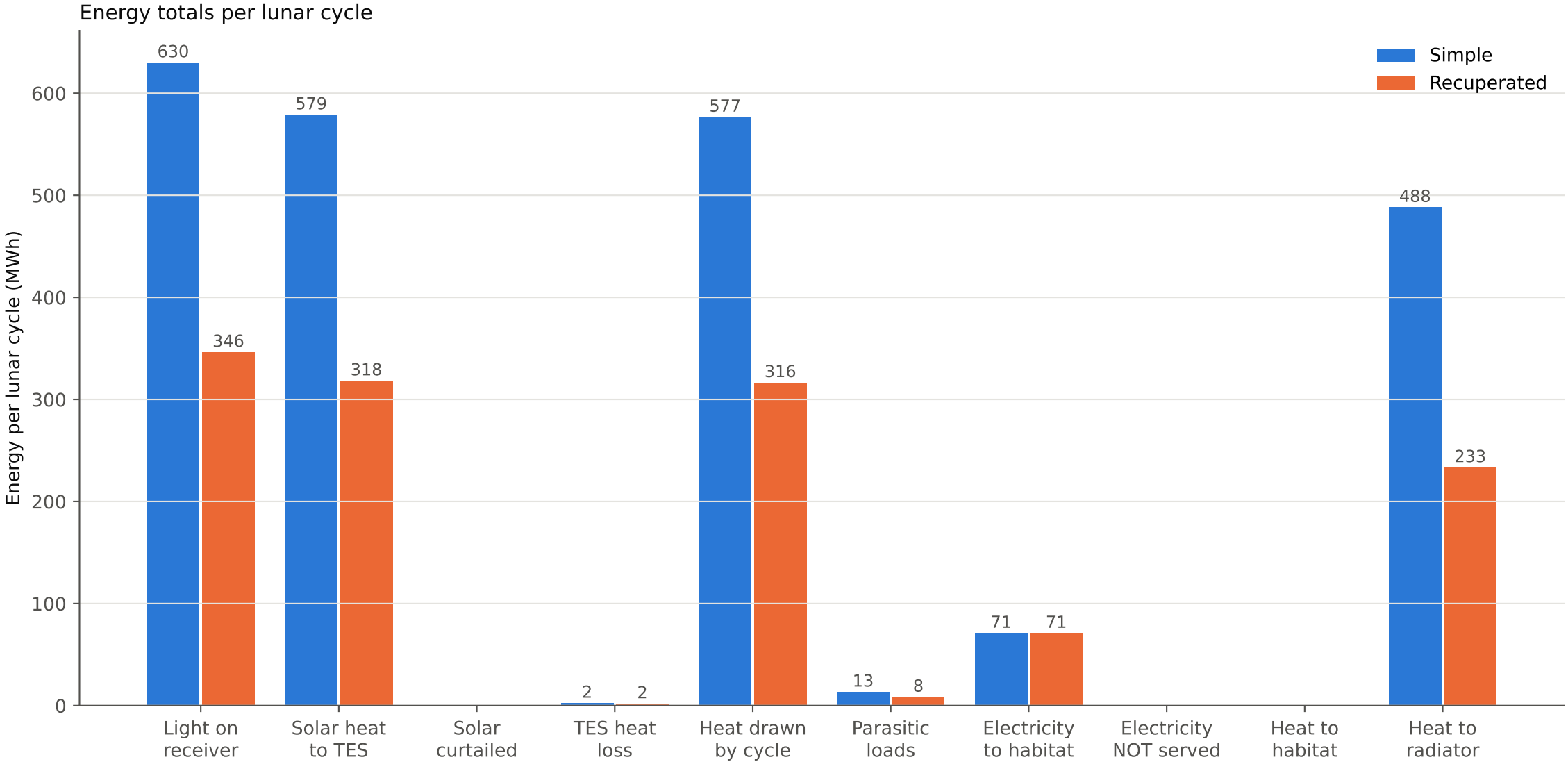

TES temperature over one lunar cycle (dots = lowest point)
Simple
Recuperated
TES maximum (800 K)
TES minimum (450 K)
Temperature (K)
800
750
700
650
600
550
500
450
0
5
10
15
20
25
30
Time since sunrise (Earth days) [shaded = lunar night]

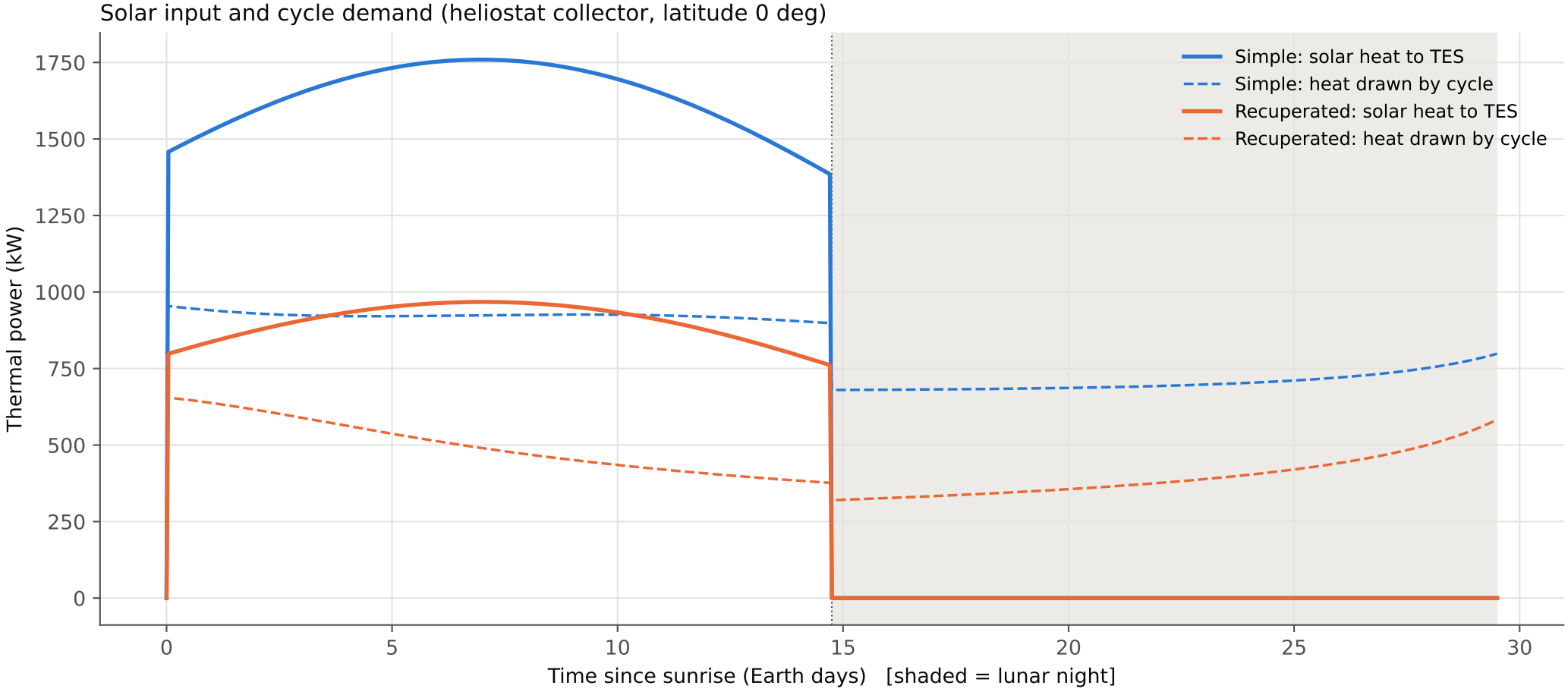
Plant performance as the TES discharges



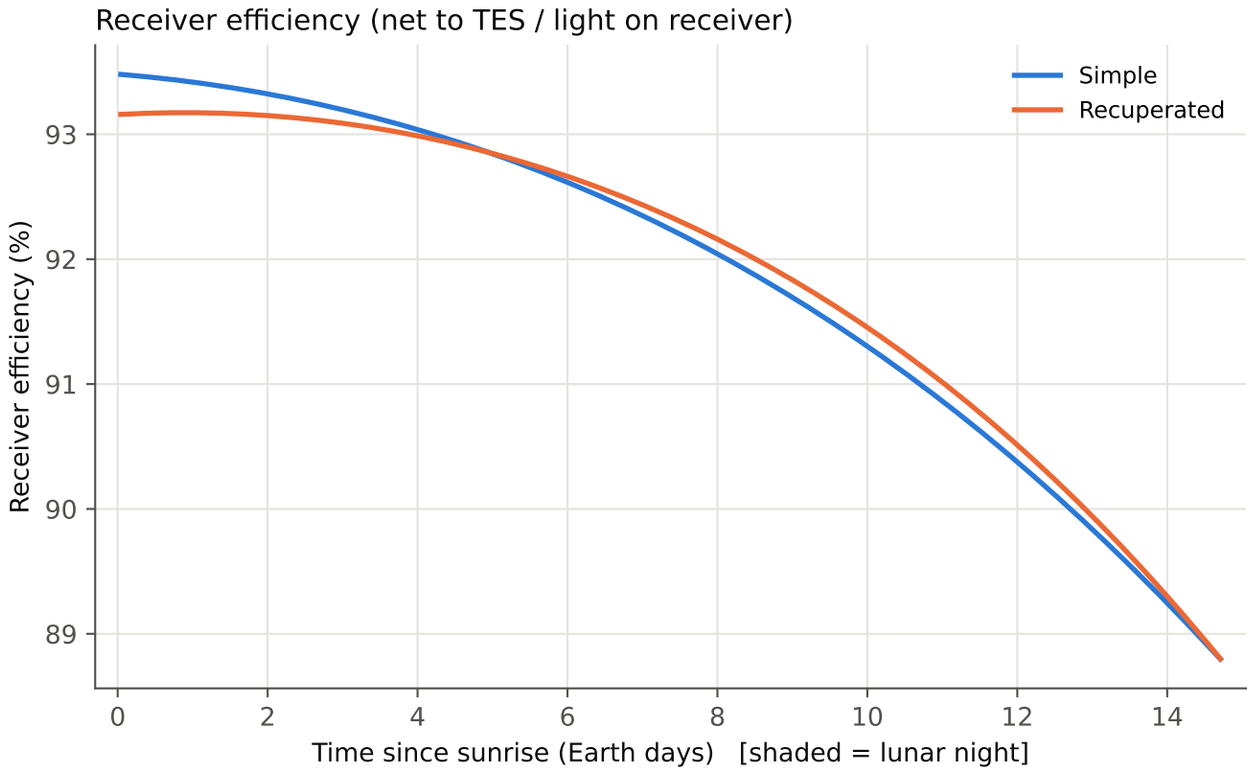



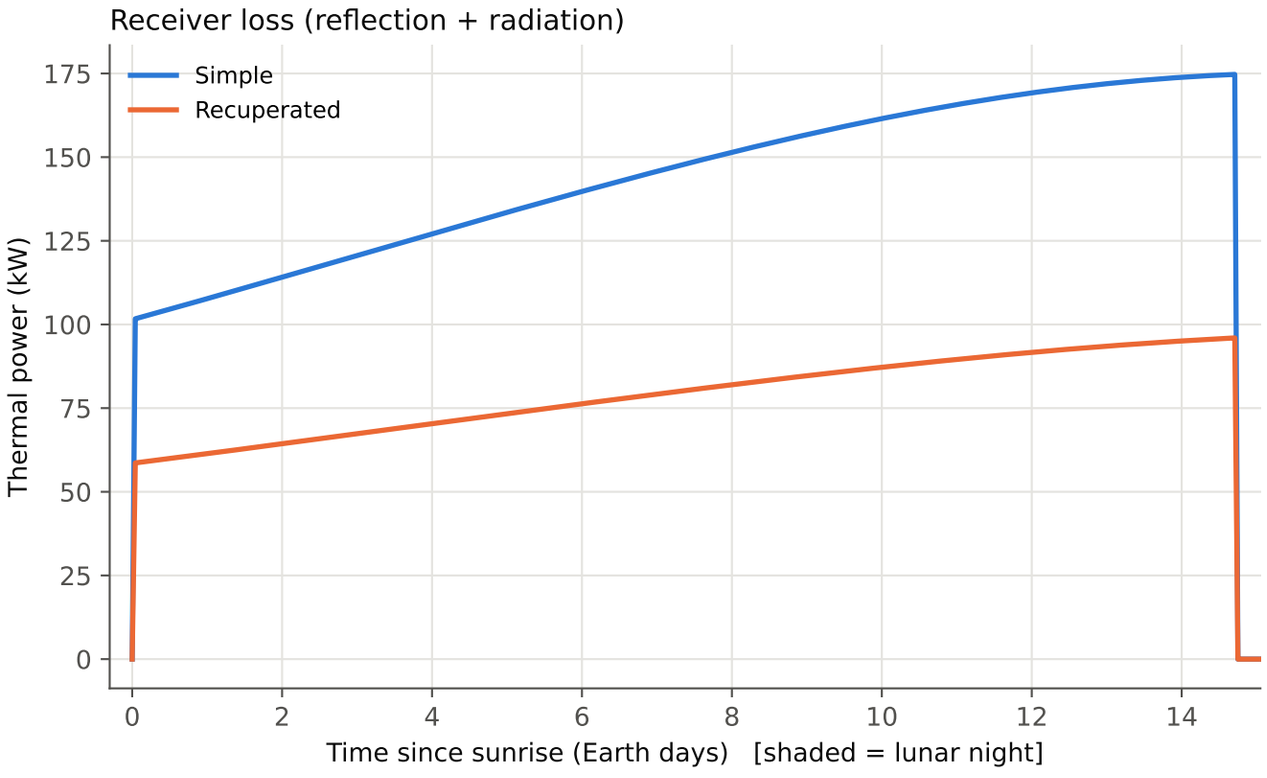

T-s diagram at full charge (TIT 790 K)
CO2 saturation dome
Simple: eta 15.9%
Recuperated: eta 33.4%
Recuperator (5-6 heats 2-3)
Temperature (K)
Entropy (kJ/kg-K)
800
700
600
500
400
300
200
0.5
1.0
1.5
2.0
2.5
1
2
3
4
5
6


Energy totals per lunar cycle
Energy per lunar cycle (MWh)
0
100
200
300
400
500
600
Simple
Recuperated
630
346
579
318
577
316
2
2
13
8
71
71
488
233
Light on receiver
Solar heat to TES
Solar curtailed
TES heat loss
Heat drawn by cycle
Parasitic loads
Electricity to habitat
Electricity NOT served
Heat to habitat
Heat to radiator

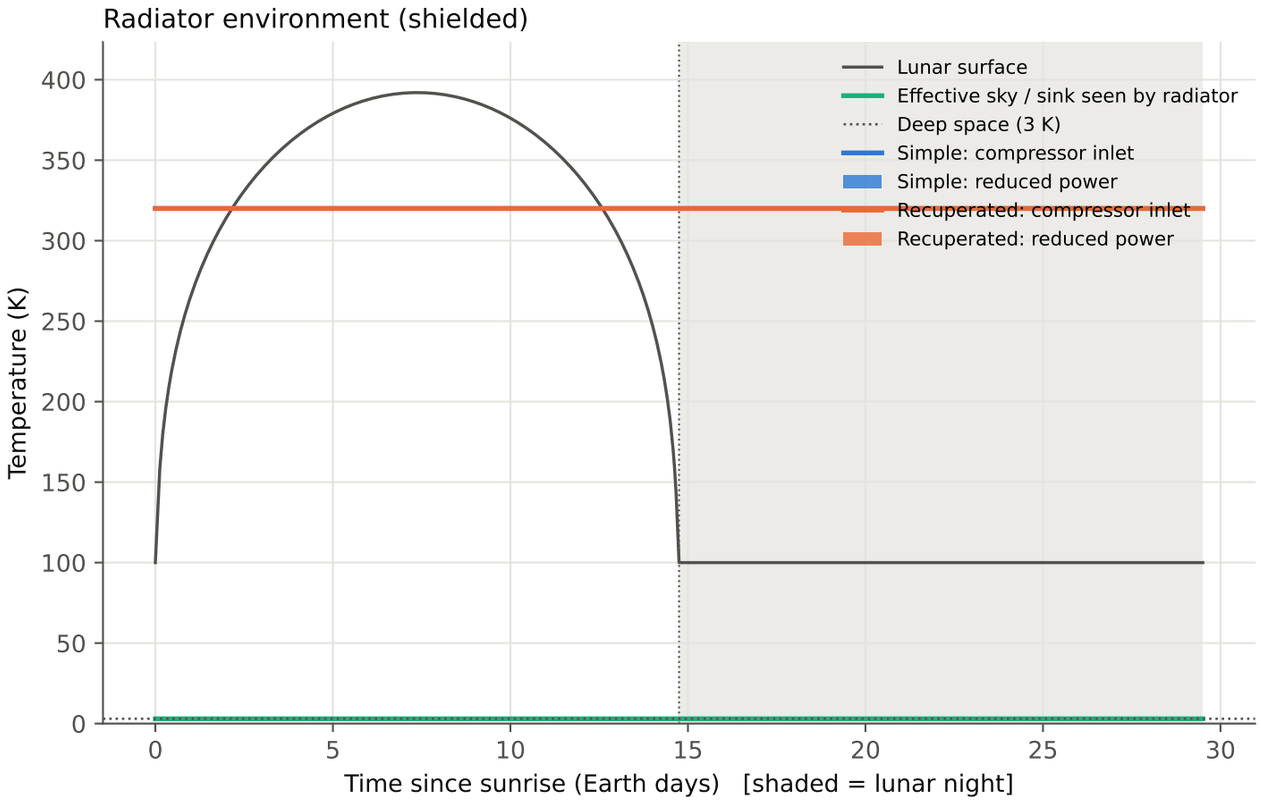

Solar input and cycle demand (heliostat collector, latitude 0 deg)
Thermal power (kW)
1750
1500
1250
1000
750
500
250
0
0
5
10
15
20
25
30
Time since sunrise (Earth days)  [shaded = lunar night]
Simple: solar heat to TES
Simple: heat drawn by cycle
Recuperated: solar heat to TES
Recuperated: heat drawn by cycle

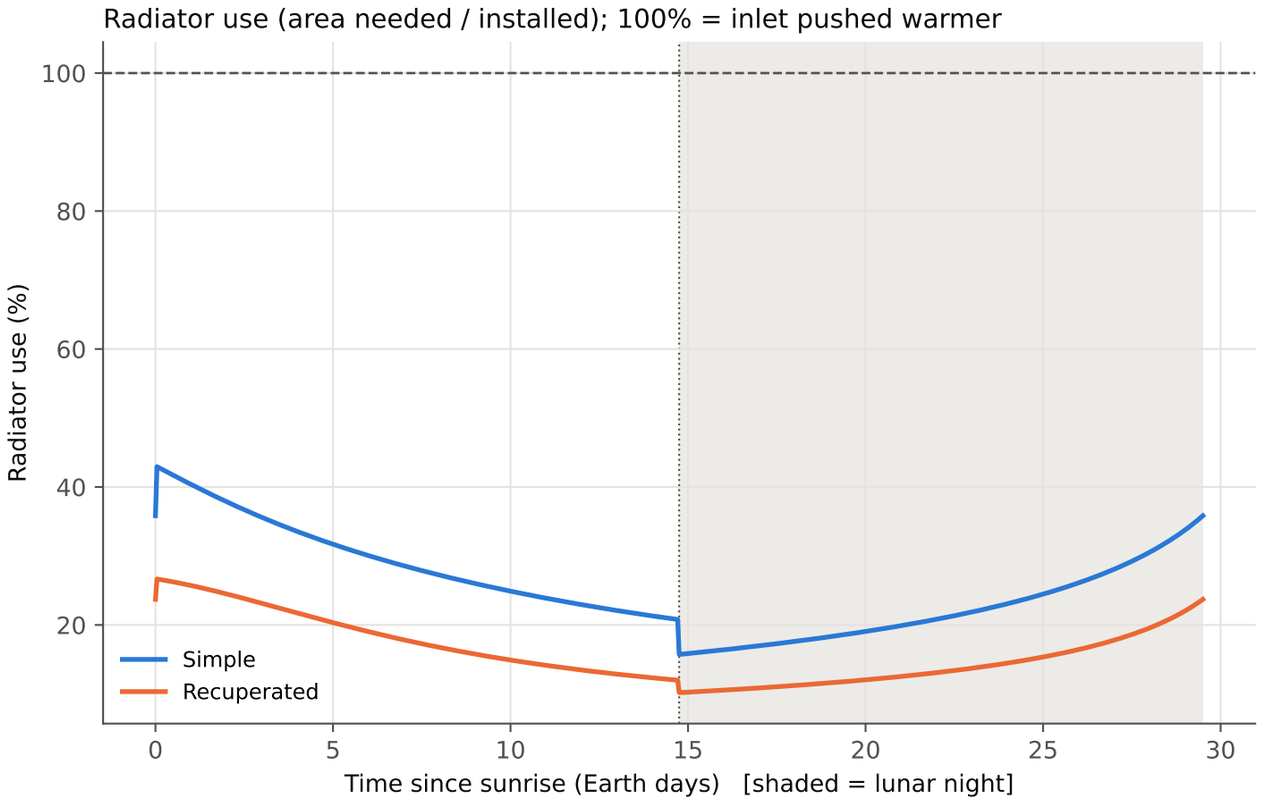
Receiver: 500x concentration, emissivity 0.85, T_rec = T_TES + 50 K

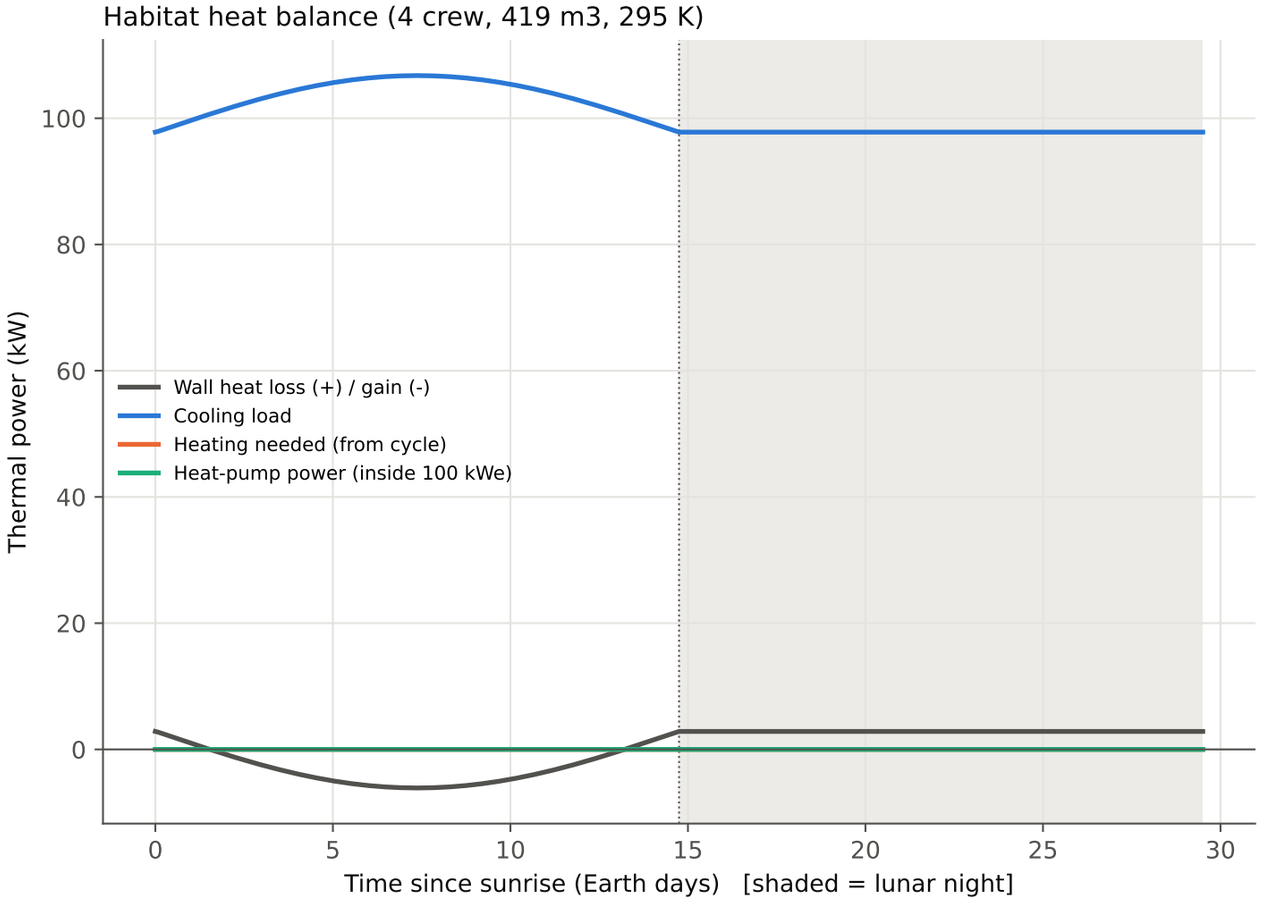


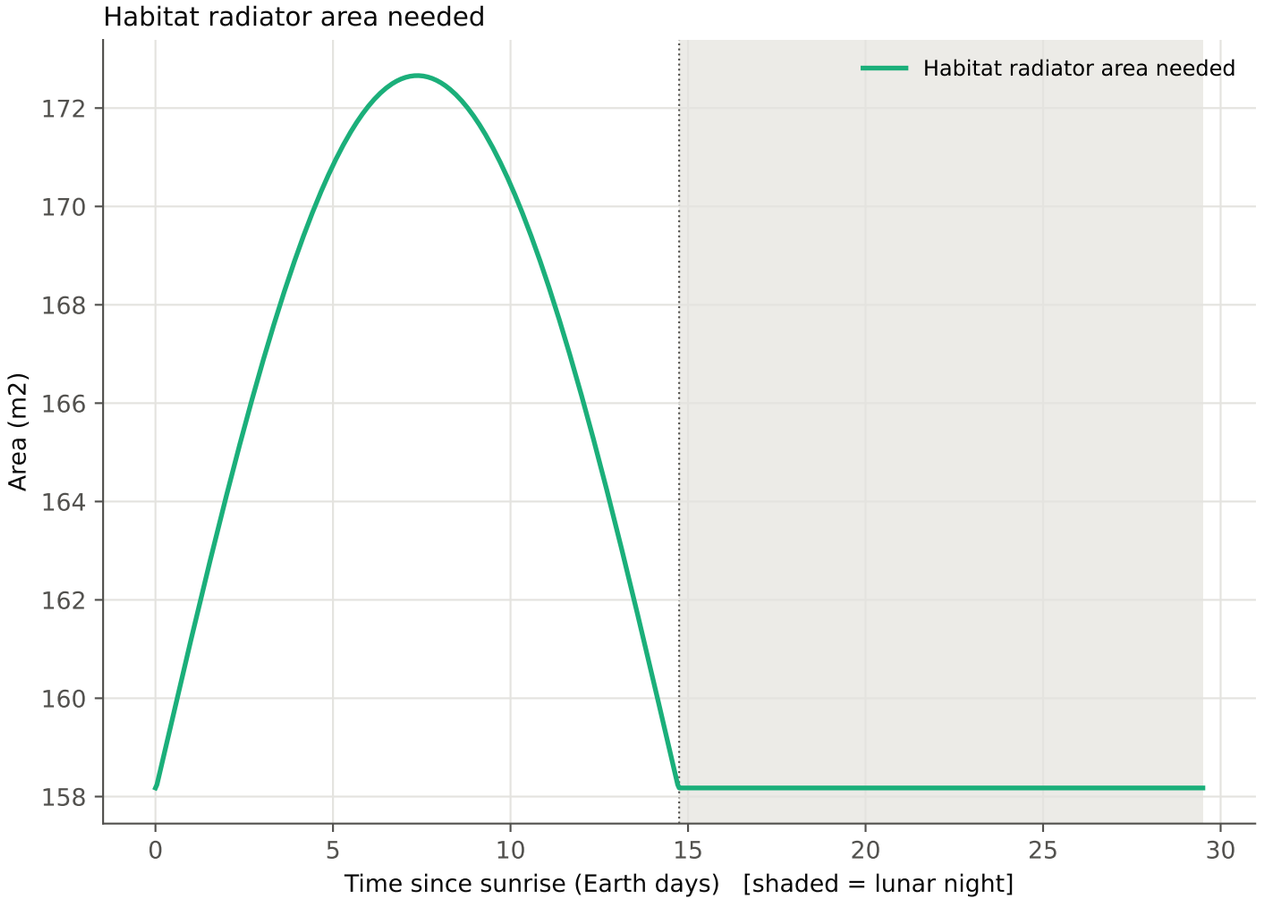



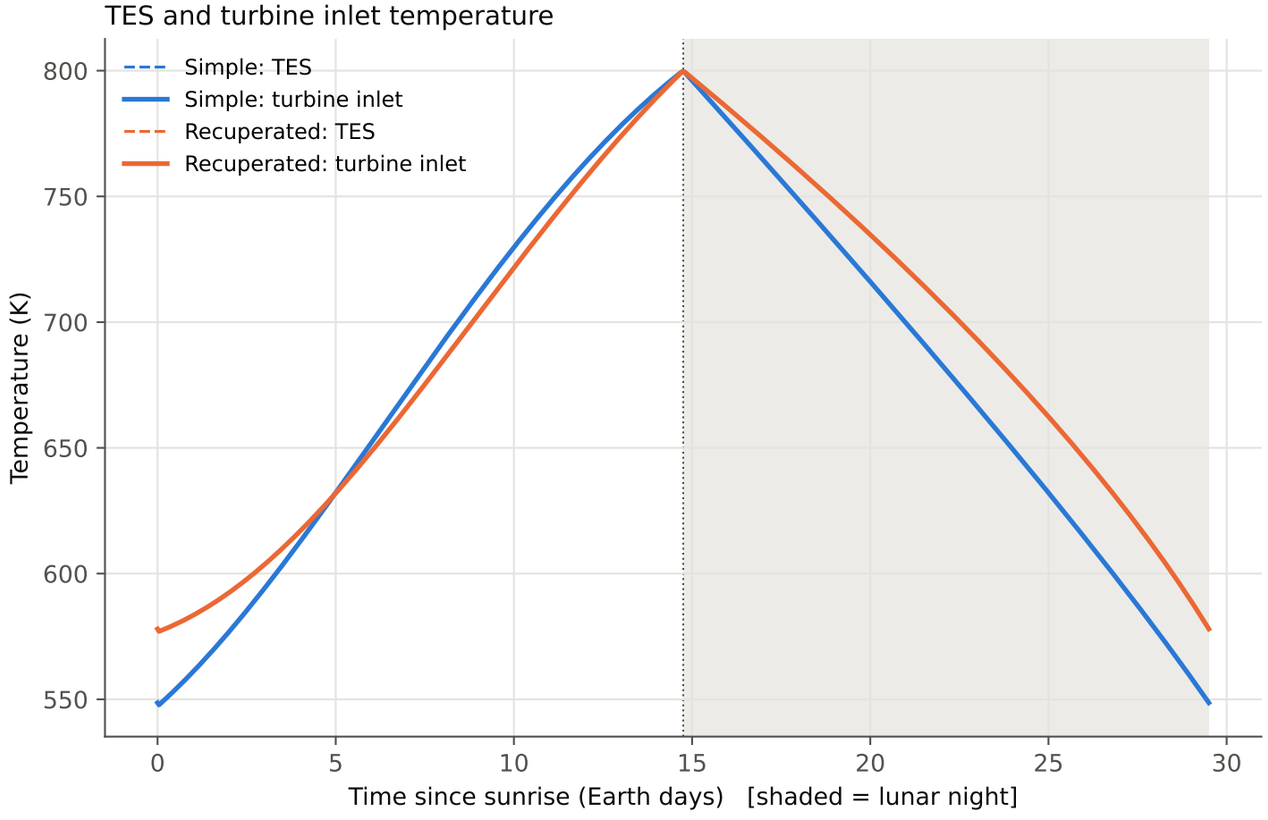
Installed radiator: Simple 1128 m2, Recuperated 1284 m2

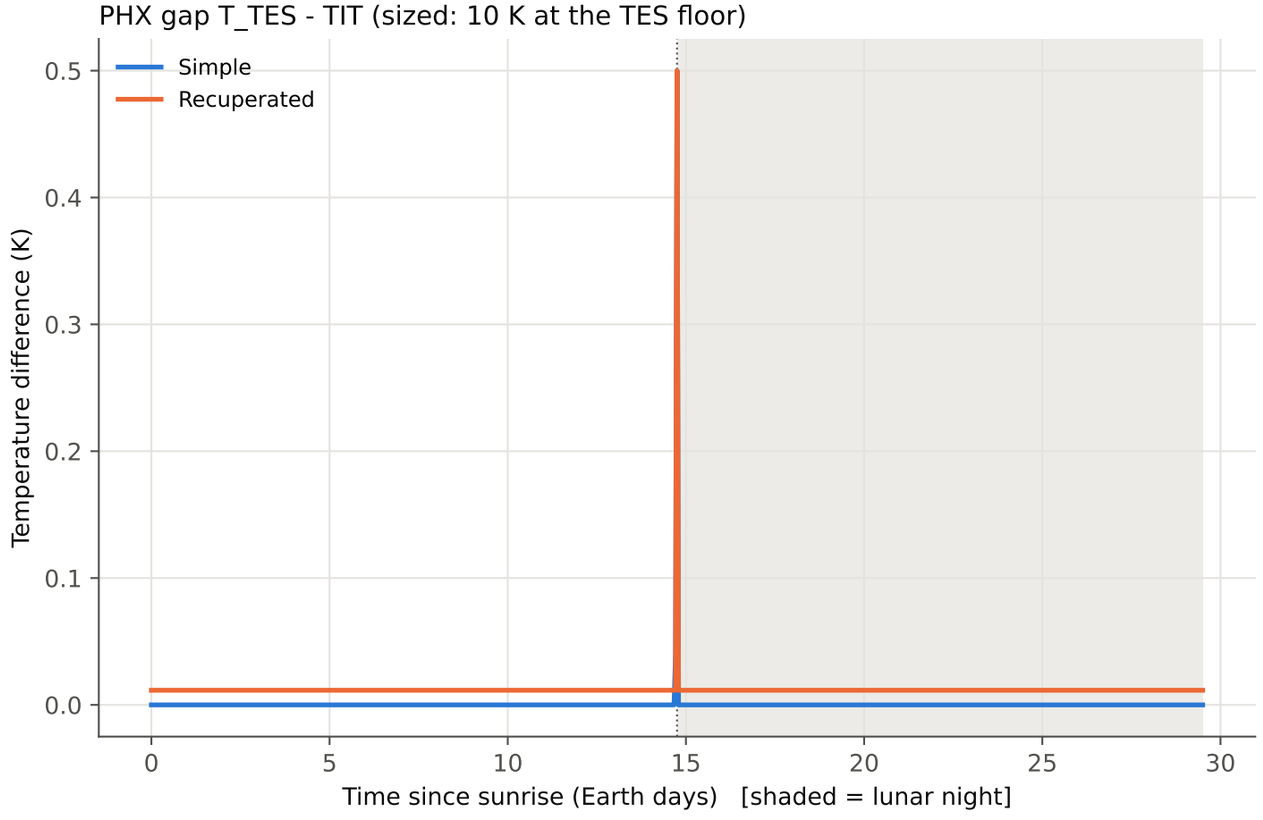



Habitat heat balance (4 crew, 419 m3, 295 K)
Thermal power (kW)
100
80
60
40
20
0
Wall heat loss (+) / gain (-)
Cooling load
Heating needed (from cycle)
Heat-pump power (inside 100 kWe)
0
5
10
15
20
25
30
Time since sunrise (Earth days)  [shaded = lunar night]


Habitat radiator area needed
Area (m2)
172
170
168
166
164
162
160
158
Habitat radiator area needed
0
5
10
15
20
25
30
Time since sunrise (Earth days)  [shaded = lunar night]


TES and turbine inlet temperature
Simple: TES
Simple: turbine inlet
Recuperated: TES
Recuperated: turbine inlet
Temperature (K)
800
750
700
650
600
550
0
5
10
15
20
25
30
Time since sunrise (Earth days) [shaded = lunar night]


PHX gap T_TES - TIT (sized: 10 K at the TES floor)
Simple
Recuperated
Temperature difference (K)
0.5
0.4
0.3
0.2
0.1
0.0
0
5
10
15
20
25
30
Time since sunrise (Earth days) [shaded = lunar night]

# References for the input values (Step 14)

REFERENCES FOR INPUT VALUES
==========


[R1] G_sun = 1361 W/m2
Kopp, G., Lean, J.L. (2011). A new, lower value of total solar irradiance: Evidence and climate significance. Geophysical Research Letters 38, L01706. doi:10.1029/2010GL045777

[R2] rho_mirror = 0.95
Typical silvered-glass heliostat reflectance (~95 %); assumed, see e.g. Stine, W.B., Geyer, M. (2001). Power From The Sun, Ch. 10 (central receivers)

[R3] f_dust = 0.93
Terrestrial heliostat soiling losses of 6.3-8.8 % (Strachan et al., 1993, Sandia), as summarised in Guan, Z. et al., Heat Exchanger Fouling and Cleaning conf. Lunar dust loss is not known: assumed

[R4] alpha_rec, eps_rec = 0.95, 0.85
Ho, C.K., Iverson, B.D. (2014). Review of high-temperature central receiver designs for concentrating solar power. Renewable and Sustainable Energy Reviews 29, 835-846. doi:10.1016/j.rser.2013.08.099

[R5] C_ratio = 500 (towers 500-1000)
Ho & Iverson (2014) [R4]: baseline concentration ~900; central receivers typically 500-1000 suns. Ho, C.K. (2017). Advances in central receivers for concentrating solar applications. Solar Energy 152, 38-56. doi:10.1016/j.solener.2017.03.048

[R6] cosine factor (Step 8) = cos(0.5 * angle sun-receiver)
Heliostat normal bisects the sun and receiver directions; effective area reduced by the cosine of half that angle: Stine & Geyer (2001) [R2], Ch. 10

[R7] Regolith cp(T), k(T), rho_TES = Table 3 correlations, 1500 kg/m3
Paper Table 3, ref. [16]: Schreiner, S.S., Dominguez, J.A., Sibille, L., Hoffman, J.A. (2016). Thermophysical property models for lunar regolith. Advances in Space Research 57(5), 1209-1222 -- check volume/pages/DOI against the paper's ref. [16]

[R8] T_surf_noon, T_surf_night = 392 K, 100 K
Vasavada, A.R. et al. (2012). Lunar equatorial surface temperatures and regolith properties from the Diviner Lunar Radiometer Experiment. JGR Planets 117, E00H18. doi:10.1029/2011JE003987 (385 K noon, 101 K midnight, 95 K pre-dawn); Williams, J.-P. et al. (2017). The global surface temperatures of the Moon as measured by Diviner. Icarus 283, 300-325. doi:10.1016/j.icarus.2016.08.012 (387-397 K at the equator)

[R9] T_ground = 250 K
Hayne, P.O. et al. (2017). Global regolith thermophysical properties of the Moon from Diviner. JGR Planets 122, 2371-2400. doi:10.1002/2017JE005387 (diurnal mean ~252-256 K at ~1 m depth, from Keihm et al. 1973 Apollo heat-flow data)

[R10] eta_comp, eta_turb = 0.85, 0.90
NETL (2014). An Assessment of Supercritical CO2 Power Cycles Integrated with Generic Heat Sources, DOE/NETL-4001/09/091014 (compressor 0.85); turbine 0.90 is a common sCO2 assumption (e.g. TESPy sCO2 recompression model, tespy.readthedocs.io)

[R11] eps_recup = 0.95
Effect of actual recuperators' effectiveness on the attainable efficiency of sCO2 Brayton cycles for solar thermal power plants. Energies 15 (2022) 7773. doi:10.3390/en15207773 (effectiveness 0.95 assumed for recuperators)

[R12] rad_sides, rad_areal_mass, T_hab_rad = 2-sided, 14 kg/m2, 283 K
Schunk, R.G. et al. (2022). Thermal Control System Architecture and Technology Challenges for a Lunar Surface Habitat. IEEE Aerospace Conf., NTRS 20210026557 (48 m2 double-sided radiators, ISS HRS areal density 14 kg/m2); Schunk et al., TFAWS 2021, NTRS 20210020015 (radiator temperature 283 K)

[R13] V_hab, n_crew = 419 m3, 4 crew
Kennedy, K.J. et al. (2008). Lunar Architecture Team Phase 2 Habitat Volume Estimation: 'Caution When Using Analogs'. Earth & Space 2008, ASCE. NTRS 20080013517 (mean estimated total pressurized volume 419 m3 for 4 crew)

[R14] eps_MLI = 0.02
MLI effective emissivity typically 0.015-0.03: Applied Sciences 12 (2022) 2858. doi:10.3390/app12062858

[R15] f_carnot_hp, T_hp_cond = 0.43, 360 K
Goldman, J., Lovell, T., Ewert, M., Petete, P. (1994). High Temperature Lift Heat Pump Refrigerant and Thermodynamic Cycle Selection. SAE 941272 (COP 43 % of Carnot; optimum rejection 377-387 K); Sridhar & Gottmann, NASA CR-190063 (1992) (radiator ~362 K)

[R16] q_metab = 14 MJ/person/day
A More Completely Defined CELSS. SAE Technical Paper 941292 (1994) (metabolic load 14 MJ/person/day)

[R17] eta_gen, T_comp_in range, P ranges = 0.95, 305-320 K, 7.5-8.0 / 20-25 MPa
Paper Table 2

[R19] w_helio_drive = 0.5 W/m2
Heliostat drive power: commercial fields have used ~0.5-4.5 % of plant output for tracking drives (US Patent 9,170,032, heliostat tracking and operation); 0.5 W/m2 is an assumed low-power value for small lunar heliostats

[R18] Recuperator/PHX heat transfer (context for dT_PHX) = U ~ 300-400 W/m2-K

## References for the input values (Step 14)  (continued)

Albrecht, K.J., Ho, C.K. et al. (Sandia), Development and testing of a 20 kWt moving packed-bed particle-to-sCO2 heat exchanger, OSTI 1883488

ASSUMED VALUES (state in the paper, cover with sensitivity)

| | |
|---|---|
| dT_rec = 50 K | receiver above the TES; heat must soak into the store |
| f_intercept = 0.95, beta_target_deg = 30 | heliostat spillage/blocking, field geometry |
| dT_PHX = 10 K at the lowest usable TES temperature | PHX size (UA) for full power down to the floor |
| dp_PHX, dp_recup, dp_cooler = 1 % | component pressure drops |
| W_aux_fixed = 3 kWe | controls, sensors, power electronics |
| dp_charge = 2 %, dT_charge = 150 K, eta_circ = 0.7 | day charging loop circulator |
| TES_mass_margin = 1.20, rad_area_margin = 1.00 | design margins |
| L_ins = 1 m | loose-regolith insulation around the TES |
| D_hab = 6 m, f_elec_inside = 1.0 | habitat shape; all 100 kWe dissipated inside |
| shield = True, F_rad_surf = 0.5 | radiators shaded from the lunar surface |
| T_comp_in_max = 380 K | highest compressor inlet before reduced power |